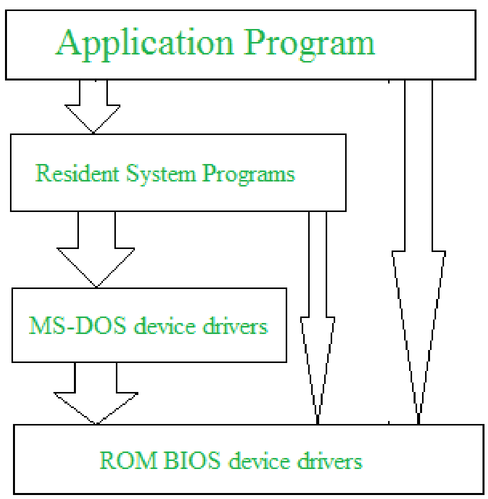
Unit 2: Operating System Structure

Operating system can be implemented with the help of various structures. The structure of the OS depends mainly on how the various common components of the operating system are interconnected and melded into the kernel. Depending on this we have following structures of the operating system: simple structure, monolithic systems, layered systems, microkernels, client-server systems, virtual machines, and exokernels.

Simple structure:

Such operating systems do not have well defined structure and are small, simple and limited systems. The interfaces and levels of functionality are not well separated. MS-DOS is an example of such operating system. These types of operating system cause the entire system to crash if one of the user programs fails.

Diagram of the structure of MS-DOS is shown below:



Advantages of Simple structure:

1. It delivers better application performance because of the few interfaces between the application program and the hardware.
2. Easy for kernel developers to develop such an operating system.

Disadvantages of Simple structure:

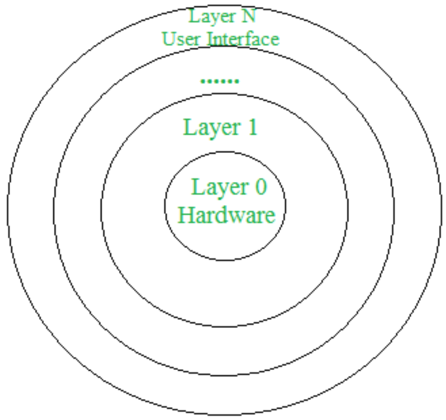
1. The structure is very complicated as no clear boundaries exists between modules.
2. It does not enforce data hiding in the operating system.

Layered System:

This approach divides the Operating System in to multiple layers. The top most layer (layer N) is the user interface and the bottom most layer (layer 0) is the hardware layer.

These layers are so designed that each layer uses the functions of the lower level layers only.

The main disadvantage of this structure is that at each layer, the data needs to be modified and passed on which adds overhead to the system. Moreover careful planning of the layers is necessary as a layer can use only lower level layers. UNIX is an example of this structure.



Advantages of Layered structure:

1. Layering makes it easier to enhance the operating system as implementation of a layer can be changed easily without affecting the other layers.
2. It is very easy to perform debugging and system verification.

Disadvantages of Layered structure:

1. In this structure the application performance is degraded as compared to simple structure.
2. It requires careful planning for designing the layers as higher layers use the functionalities of only the lower layers.

Kernel

The kernel is a computer program at the core of a computer's operating system with complete control over everything in the system. It is an integral part of any operating system. It is the "portion of the operating system code that is always resident in memory"

A Kernel is a computer program that is the heart and core of an Operating System. Since the Operating System has control over the system so, the Kernel also has control over everything in the system. It is the most important part of an Operating System.

Whenever a system starts, the Kernel is the first program that is loaded after the bootloader because the Kernel has to handle the rest of the thing of the system for the Operating System. The Kernel remains in the memory until the Operating System is shut-down.

The Kernel is responsible for low-level tasks such as disk management, memory management, task management, etc. It provides an interface between the user and the hardware components of the system. When a process makes a request to the Kernel, then it is called System Call.

A Kernel is provided with a protected Kernel Space which is a separate area of memory and this area is not accessible by other application programs. So, the code of the Kernel is loaded into this protected Kernel Space. Apart from this, the memory used by other applications is called the User Space. As these are two different spaces in the memory, so communication between them is a bit slower.

Types of Kernel: Monolithic kernel and Microkernel.

Functions of a Kernel

Following are the functions of a Kernel:

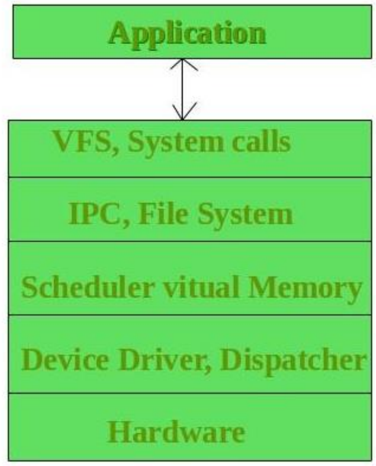
- **Access Computer resource:** A Kernel can access various computer resources like the CPU, I/O devices and other resources. It acts as a bridge between the user and the resources of the system.
- **Resource Management:** It is the duty of a Kernel to share the resources between various process in such a way that there is uniform access to the resources by every process.
- **Memory Management:** Every process needs some memory space. So, memory must be allocated and deallocated for its execution. All these memory management is done by a Kernel.
- **Device Management:** The peripheral devices connected in the system are used by the processes. So, the allocation of these devices is managed by the Kernel.
- **Interrupt Handling:** The interrupts from hardware/software are detected and handled by the Kernel.

SN	Key	Operating System	Kernel
1	Types	Operating system is a system software.	Kernel is a part of operating system.
2	Work	Operating system acts as an interface between user and hardware.	Kernel acts as an interface between applications and hardware.
3	Main tasks	Ease of doing system operations, security etc.	Memory management, space management, process management and task management.
4	Basis	A computer need Operating System to run.	An Operating System needs Kernel to run.
5	Types	Operating Systems types are multiuser, multitasking, multiprocessor, realtime, distributed etc.	Kernel types are monolithic kernel and micro kernel.
6	Boot	Operating System is the first program to load when computer boots up.	Kernel is the first program to load when operating system loads.

Monolithic kernel

In monolithic kernel, user services and kernel services are implemented under same address space. It increases the size of the kernel, thus increases size of operating system as well.

This kernel provides CPU scheduling, memory management, file management and other operating system functions through system calls. As both services are implemented under same address space, this makes operating system execution faster.



Monolithic Kernel

If any service fails the entire system crashes, and it is one of the drawbacks of this kernel. The entire operating system needs modification if user adds a new service.
Example of some Monolithic Kernel based OSs are: Unix, Linux, Open VMS, XTS-400, z/TPF.

Advantages of Monolithic Kernel –

1. It provides CPU scheduling, memory management, file management and other operating system functions through system calls.
2. It is a single large process running entirely in a single address space.
3. It is a single static binary file.

Disadvantages of Monolithic Kernel –

1. If anyone service fails it leads to entire system failure.
2. If user has to add any new service. User needs to modify entire operating system.

Micro-kernel:

This structure designs the operating system by removing all non-essential components from the kernel and implementing them as system and user programs. This result in a smaller kernel called the micro-kernel.
Advantages of this structure are that all new services need to be added to user space and does not require the kernel to be modified. Thus it is more secure and reliable as if a service fails then rest of the operating system remains untouched. Mac OS is an example of this type of OS.

Advantages of Micro-kernel structure:

1. All new services need to be added to user space and does not require the kernel to be modified.
2. It makes the operating system portable to various platforms.
3. As microkernels are small so these can be tested effectively.

Disadvantages of Micro-kernel structure:

1. Increased level of inter module communication degrades system performance.

SN	Key	Monolithic Kernel	Microkernel
1	Size	It is larger in size.	It is smaller in size.
2	Execution	It is faster for execution.	It is slower for execution.
3	Extendibility	It is complex to extend	It is easier to extend.
4	Security	If a service crashes, whole system will crash	It a service crashes, it will not affect the working of microkernel
5	Code	Less code is required to write monolithic kernel	More code is required to write microkernel
6	Database	It has shared database	Each module has its own database

Hybrid Kernel

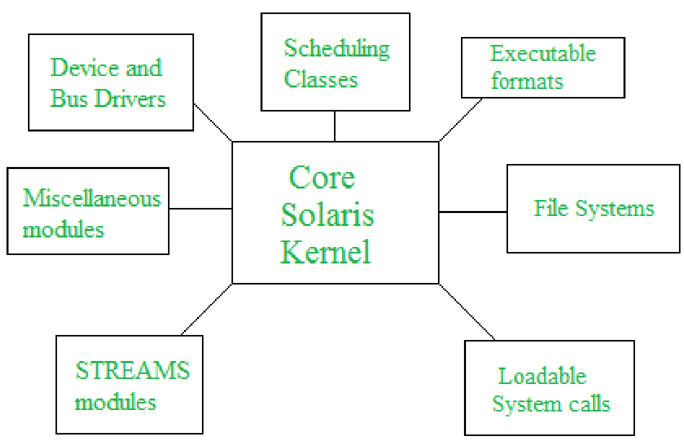
It is an operating system kernel architecture that attempts to combine aspects and benefits of microkernel and monolithic kernel architectures.
It makes use of the speed of the monolithic kernel and modularity of microkernel. It uses some services like network stack or file system in kernel space to enhance performance while other non-essential services from the user space.

Advantages of Hybrid kernels:
1. Faster deployment time for drivers that can operate within modules.
2. Integration of best features of monolithic and microkernel.
3. Faster integration of third party technology.

Disadvantages of Hybrid kernels:
1. With more interfaces to pass through, the possibility of increased bug exists.
2. Maintaining modules can be confusing for some administrators.

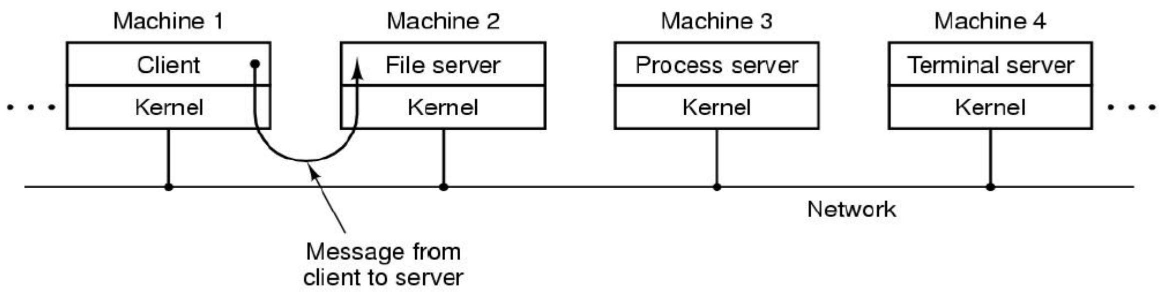
Modular structure or approach:

It is considered as the best approach for an OS. It involves designing of a modular kernel. The kernel has only set of core components and other services are added as dynamically loadable modules to the kernel either during run time or boot time. It resembles layered structure due to the fact that each kernel has defined and protected interfaces but it is more flexible than the layered structure as a module can call any other module. Solaris OS is an example that uses this approach.



Client-Server Model

It is a variation of the microkernel idea is to distinguish two classes of processes, the servers, each of which provides some service, and the **clients,** which use these services.
It is a distributed application structure that partitions tasks or workloads between the provider of a resource (server) and server requesters (client).



Communication between clients and servers is often by message passing. To obtain a service, a client process constructs a message saying what it wants and sends it to the appropriate service. The service then does the work and sends back the answer. If the client and server run on the same machine, certain optimizations are possible, but conceptually, we are talking about message passing here.

Advantages of Client-Server model:

1. Centralized system with all data in a single place.
2. Cost efficient requires less maintenance cost and Data recovery is possible.
3. The capacity of client and servers can be changed separately.

Disadvantages of Client-Server model:

1. Client is prone to virus, worm or Trojan if the server is infected.
2. Server is prone to Denial of Service attack.
3. Data packets may be spoofed or modified during transmission.

Virtual Machines

A virtual machine (VM) is an operating system (OS) or application environment that is installed on software, which imitates dedicated hardware. The end user has the same experience on a virtual machine as they would have on dedicated hardware.

Specialized software, called a hypervisor, emulates the PC client or server's CPU, memory, hard disk, network and other hardware resources completely, enabling virtual machines to share the resources.

The hypervisor can emulate multiple virtual hardware platforms that are isolated from each other, allowing virtual machines to run Linux and Windows Server operating systems on the same underlying physical host. Virtualization limits costs by reducing the need for physical hardware systems.

Virtual machines more efficiently use hardware, which lowers the quantities of hardware and associated maintenance costs, and reduces power and cooling demand. They also ease management because virtual hardware does not fail.

Administrators can take advantage of virtual environments to simplify backups, disaster recovery, new deployments and basic system administration tasks.

Uses for Virtual Machines

Test new versions of operating systems: You can try out Windows 10 on your Windows 7 computer if you aren't willing to upgrade yet.

Experiment with other operating systems: Installing various distributions of Linux in a virtual machine lets you experiment with them and learn how they work. And running macOS on Windows 10 in a virtual machine lets you get used to a different operating system you're considering using full-time.

Run software designed for another operating systems: Mac and Linux users can run Windows in a virtual machine to use Windows software on their computers without compatibility headache. Unfortunately, games are a problem. Virtual machine programs introduce overhead and 3D games will not run smoothly in a VM.

Test software on multiple platforms: If you need to test whether an application works on multiple operating systems, you can install each in a virtual machine.

Consolidate servers: For businesses running multiple servers, they can place some into virtual machines and run them on a single computer. Each virtual machine is an isolated container, so this doesn't introduce the security headaches involved with running different servers on the same operating system. The virtual machines can also be moved between physical servers.

Exokernels

Exokernel is an operating system developed at the MIT that provides application-level management of hardware resources. This architecture is designed to separate resource protection from management to facilitate application-specific customization.

Some of the features of exokernel operating systems include:

- Better support for application control
- Separates security from management
- Abstractions are moved securely to an untrusted library operating system
- Provides a low-level interface
- Library operating systems offer portability and compatibility

Principles of Exokernels

1. Separate protection and management: Resource management is restricted to functions necessary for protection.

2. Expose allocation: Applications allocate resources explicitly.

3. Expose name: Exokernels use physical names wherever possible.

4. Expose revocation: Exokernels let applications to choose which instance of a resource to give up.

5. Expose information: Exokernels expose all system information and collect data that applications cannot easily derive locally.

Advantages of Exokernels:

1. Significant performance increase.
2. Applications can make more efficient and intelligent use of hardware resources by being aware of resource availability, revocation and allocation.
3. Ease development and testing of new operating system ideas. (New scheduling techniques, memory management methods, etc)

Disadvantages of Exokernels:

1. Complexity in design of exokernel interfaces.
2. Less consistency.

Shell

A shell is a software that provides an interface between users and operating system of a computer to access the services of a kernel. Shell accept human readable commands from user and convert them into something which kernel can understand. It is a command language interpreter that execute commands read from input devices such as keyboards or from files. The shell gets started when the user logs in or start the terminal.

Shell is broadly classified into two categories –Command Line Shell and Graphical shell

Command Line Shell

Shell can be accessed by user using a command line interface. A special program called Terminal in linux/macOS or Command Prompt in Windows OS is provided to type in the human readable commands such as “cat”, “ls” etc. and then it is being execute.

Graphical Shells

Graphical shells provide means for manipulating programs based on graphical user interface (GUI), by allowing for operations such as opening, closing, moving and resizing windows, as well as switching focus between windows. Window OS or Ubuntu OS can be considered as good example which provide GUI to user for interacting with program. User do not need to type in command for every actions.